

NARRANDERA, Australia

WMO: 957060

Lat: 34.7050S

Lon: 146.5139E

Elev: 146

StdP: 99.59

Time Zone: 10.00 (AUS)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.9	0.4	-4.3	2.7	20.7	-2.9	3.0	16.4	10.5	15.0	9.3	13.4	0.4	140	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	39.8	19.5	37.3	19.2	35.2	18.7	22.2	30.4	21.3	30.0	20.7	29.7	5.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.4	23.8	19.2	14.2	23.7	18.0	13.2	23.3	65.7	30.4	62.6	29.1	60.3	29.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.7	7.8	-2.9	42.8	1.2	1.9	-3.7	44.2	-4.4	45.4	-5.1	46.5	-6.0	47.9		
			-3.2	23.6	1.3	1.2	-4.1	24.5	-4.9	25.2	-5.6	25.8	-6.5	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	26.0	24.9	22.0	17.2	12.5	9.6	9.0	9.7	12.8	17.4	21.0	23.4	(6)
(7)		DBStd	6.99	4.08	3.94	3.96	3.61	2.81	2.50	2.36	2.59	3.18	4.23	4.26	4.14	(7)
(8)		HDD10.0	137	0	0	0	1	7	38	49	37	6	0	0	0	(8)
(9)		HDD18.3	1348	1	2	13	63	183	263	291	268	170	72	19	5	(9)
(10)		CDD10.0	2717	497	416	371	218	84	24	17	27	89	228	329	415	(10)
(11)		CDD18.3	884	239	184	125	29	1	0	0	0	4	42	99	161	(11)
(12)		CDH23.3	11473	3259	2232	1368	327	9	0	0	3	83	652	1335	2205	(12)
(13)		CDH26.7	5870	1907	1182	618	84	0	0	0	0	19	257	648	1154	(13)
(14)	Wind	WSAvg	3.4	4.1	3.9	3.6	2.8	2.7	2.6	3.0	3.1	3.5	3.7	4.0	4.0	(14)
(15)	Precipitation	PrecAvg	420	35	34	29	36	42	37	37	41	37	42	31	32	(15)
(16)		PrecMax	776	156	120	155	170	152	118	83	83	99	133	184	101	(16)
(17)		PrecMin	171	2	0	0	1	5	2	2	1	9	0	0	0	(17)
(18)		PrecStd	134	33	35	31	35	33	24	23	23	23	31	32	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	42.0	37.8	31.7	24.8	19.0	19.2	22.2	29.4	34.3	38.8	40.9	(19)
(20)		MCWB	20.3	20.5	19.4	17.1	15.4	12.9	12.3	13.0	14.6	16.4	19.2	19.0	19.0	(20)
(21)		2%	DB	40.2	38.5	34.5	29.0	22.7	17.3	17.0	19.9	25.5	32.0	35.3	37.6	(21)
(22)		MCWB	19.8	19.4	18.6	16.5	14.4	12.3	11.7	12.0	13.7	15.7	17.8	18.9	18.9	(22)
(23)		5%	DB	37.8	35.6	32.2	27.1	20.8	16.0	15.5	17.6	23.0	29.7	33.1	35.2	(23)
(24)		MCWB	19.4	19.4	18.3	16.0	13.7	11.5	11.2	11.2	13.1	15.8	17.3	18.3	18.3	(24)
(25)		10%	DB	35.4	33.1	30.1	25.0	18.8	14.7	14.1	15.9	20.6	27.1	30.4	32.9	(25)
(26)		MCWB	19.0	18.8	17.5	15.5	12.9	11.2	10.6	10.5	12.2	14.7	16.6	17.8	17.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	23.8	21.7	19.5	17.1	14.2	14.8	14.5	16.1	18.9	21.2	22.6	(27)
(28)		MCDWB	30.7	28.8	28.6	26.8	19.4	16.0	16.2	19.1	23.6	28.2	32.0	28.7	28.7	(28)
(29)		2%	WB	21.9	22.5	20.8	18.2	15.3	13.2	13.1	13.1	15.0	17.7	20.0	21.0	(29)
(30)		MCDWB	31.1	30.1	27.0	23.9	20.8	15.6	15.3	18.3	21.7	26.9	29.1	28.9	28.9	(30)
(31)		5%	WB	21.2	21.5	19.7	17.2	14.2	12.4	11.8	11.9	13.8	16.6	19.0	20.1	(31)
(32)		MCDWB	31.0	29.7	27.8	23.2	19.3	15.0	14.4	16.6	20.5	25.6	28.5	28.8	28.8	(32)
(33)		10%	WB	20.6	20.6	19.0	16.3	13.3	11.7	10.7	11.0	12.8	15.7	18.1	19.2	(33)
(34)		MCDWB	31.0	29.1	27.5	23.3	17.9	14.2	13.7	15.2	19.6	25.1	26.4	29.3	29.3	(34)
(35)	Mean Daily Temperature Range	MDBR	15.1	13.9	13.6	13.7	12.3	10.5	10.3	11.9	14.4	15.6	14.8	14.9	14.9	(35)
(36)		5% DB	MCDBR	17.8	17.1	15.9	15.7	14.2	12.3	12.2	14.8	17.9	18.7	18.1	17.5	(36)
(37)		MCWBR	5.4	5.4	5.7	6.7	7.3	7.7	7.5	8.1	8.3	7.2	5.8	5.7	5.7	(37)
(38)		5% WB	MCDBR	13.5	11.7	12.1	12.5	12.0	9.7	9.6	12.8	14.4	15.0	13.6	12.6	(38)
(39)		MCWBR	5.1	4.4	5.2	5.7	6.7	6.3	6.4	7.5	7.7	6.7	4.9	5.4	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.345	0.342	0.321	0.309	0.297	0.289	0.286	0.293	0.323	0.327	0.344	0.338	0.338	(40)
(41)		taud	2.478	2.517	2.558	2.556	2.560	2.589	2.596	2.555	2.452	2.472	2.462	2.489	2.489	(41)
(42)		Ebn at Noon	991	973	958	913	870	852	877	918	939	977	985	1002	1002	(42)
(43)		Edh at Noon	116	108	96	85	74	68	71	83	105	111	117	116	116	(43)
(44)	All-Sky Solar Radiation	RadAvg	7.79	6.86	5.79	4.28	3.09	2.38	2.62	3.56	4.99	6.33	7.15	7.86	7.86	(44)
(45)		RadStd	0.44	0.38	0.32	0.19	0.15	0.19	0.16	0.27	0.38	0.43	0.48	0.28	0.28	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.46	N/A	N/A	+1.16	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page